Exp No:10 Experiment to understand array function in Data science.

Description: Understand array function using Numpy library

PROGRAM:

```
import numpy as np
# Create an array
array = np.array([64, 36, 81, 49, 25, 16, 100, 9, 121])
print("Original Array:")
print(array)
# Square root of array elements
print("\nSquare Root:")
print(np.sqrt(array))
# Check dimensions
print("\nDimension of Array:")
print(array.ndim)
# Reshape array into 3x3
new_array = array.reshape(3, 3)
print("\nReshaped Array:")
print(new_array)
# Check dimensions after reshaping
print("\nDimension after Reshaping:")
print(new_array.ndim)
# Convert 2D array back to 1D
print("\nRavelled Array:")
print(new_array.ravel())
# Array slicing
print("\nSelected Elements:")
print(new_array[2, 1:3])
```

OUTPUT:

Original Array:

[64 36 81 49 25 16 100   9 121]

Square Root:

[8. 6. 9. 7. 5. 4. 10. 3. 11.]

Dimension of Array:

1

Reshaped Array:

[[64 36 81]

 [49 25 16]

 [100   9 121]]

Dimension after Reshaping:

2

Ravelled Array:

[64 36 81 49 25 16 100   9 121]

Selected Elements:

[9 121]